

Peter Tinashe Mundowa

Data Scientist & AI Engineer | Production ML & AI Agents | Deep Learning & Data Engineering | Full-Stack AI Systems

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer who builds and deploys production-grade AI systems, ML pipelines, and full-stack applications across multiple enterprise clients. Engineers AI agents and LLM-powered systems on complex, undocumented databases — turning raw operational data into reliable, real-time intelligence that businesses run on. Grounded in deep learning, statistical modelling, and data engineering, with hands-on delivery from data foundations through to deployed, monitored AI in production. Zindi Ambassador for Zimbabwe, IndabaX Best-Poster recipient, and ITU AI for Good—featured practitioner. BSc Data Science & Informatics completed 2026 (graduating August).

WORK EXPERIENCE

Data Scientist / AI Engineer Jan 2026 – Present

AI Creative Solutions Harare, Zimbabwe

- **Production AI Agents at Scale:** Designed and deployed AI agents on a large, fully undocumented enterprise ERP database — reverse-engineered a complete schema and relationship map through evidence-based pattern validation, reaching **95.56% join accuracy** across **225,000+ live transactions**. Agents serve real-time production intelligence, inventory traceability, and operational analytics, continuously validated against live data.
- **LLM Orchestration & Evaluation:** Built and evaluated LLM-driven agents for sales analytics, customer segmentation, and route performance using structured prompt engineering and systematic agent evaluation — improving response accuracy by **~30%** and cutting insight turnaround from days to under **5 minutes**.
- **Full-Stack Systems & APIs:** Built complete operations platforms from scratch — React 19, Express 5, PostgreSQL, Drizzle ORM — with role-based access control, real-time stock management, and clean REST APIs powering AI-driven features. Live in production for enterprise clients in Zimbabwe.
- **Data Engineering & Deployment:** Designed end-to-end ETL pipelines across SQL Server and PostgreSQL sources, cutting data-preparation time by **~50%** and manual reporting effort by **~60%**; shipped and iterated agents and applications through Git-based workflows and cloud hosting.

Revenue Assurance Intern Aug 2024 – Aug 2025

Steward Bank Harare, Zimbabwe

- **Exchange Rate Automation:** Built a Python pipeline that fetched RBZ exchange-rate PDFs daily, extracted and loaded the data into SQL Server, and surfaced it through Power BI/DAX dashboards — cutting manual capture from **4 hours** to **~15 minutes**.
- Ran SQL-based revenue reconciliations and built automation tools supporting audit, compliance, and risk workflows.

Machine Learning & AI Intern Dec 2023 – Feb 2024

Mentorness Pvt Ltd Remote

- Developed predictive ML models (salary estimation, fraud detection) in Python with Scikit-learn and TensorFlow, delivering deployable prototypes aligned to business requirements.

COMMUNITY & LEADERSHIP

Zindi Ambassador — Zimbabwe 2024 – Present

Organises data science events, coordinates ML competitions, and mentors practitioners across Zimbabwe. Hosted **“Building a Successful Career in Data & AI”** — 100+ attendees, with the 90-minute session extended by live Q&A.

Chapter Lead & Author 2024 – Present

Leads the **Right for Education (R:Ed) Zimbabwe Chapter** and publishes on African technology, entrepreneurship, and leadership. Recipient of the R:Ed **Leadership Beacon Award** and Certificate of Excellence in Leadership Support.

SELECTED PROJECTS

ZimCropGuard — Multi-Crop Disease Detection (*Best Poster, IndabaX Zimbabwe 2026*) 2026

- Hybrid **CNN + Vision Transformer** framework for multi-crop disease detection across varied field conditions. **Won Best Poster at IndabaX Zimbabwe 2026**; selected to represent Zimbabwe at **Deep Learning Indaba 2027**.

- CNN classifier (EfficientNetB0 transfer learning, PlantVillage dataset) deployed as a live **Streamlit** web application for real-time field disease detection.

Road Traffic Monitoring System

2024

- Real-time multi-class vehicle detection and tracking with **YOLOv8 + SORT** and a virtual counting line; exportable analytics designed for live traffic-management deployment.

LLM Trust & Safety — Adversarial Red-Teaming (Zindi)

2025

- Adversarial red-teaming and evaluation of large language models for trust & safety, probing failure modes and robustness; reached **0.828** on the public leaderboard.

East Africa Financial Health Prediction (Zindi)

2025

- Multi-class LightGBM model predicting financial health outcomes on class-imbalanced data across East African populations; achieved **MAE 0.1199**.

COMPETITIONS & RECOGNITION

Best Poster	IndabaX Zimbabwe 2026 — ZimCropGuard (CNN + Vision Transformer)	National
Selected	Deep Learning Indaba 2027 — representing Zimbabwe	International
Finalist	ITU AI for Good Innovation Challenge	International
3rd Place	ITU AI/ML Global Challenge — Environmental Footprint	International
3rd Place	IndabaX Zimbabwe Face Spoofing Detection Hackathon	National

EDUCATION

University of Zimbabwe	BSc Data Science and Informatics (<i>completed; graduating Aug</i>)	2022 – 2026
WorldQuant University	Applied AI Lab + Applied Data Science Lab	2023 – 2025
Aspire Institute	Aspire Leaders Program (global competitive cohort)	Jan – Mar 2025

CERTIFICATIONS

• Microsoft Azure AI Engineer Associate	2026	• DataCamp — Data Analyst & SQL Associate	2024
• Microsoft Azure Data Scientist Associate	2026	• McKinsey Forward Program	2024
• Microsoft Power BI Data Analyst Associate	2026	• YALI — Fundamentals of Responsible Leadership	2024
• DataCamp — Data Scientist & Data Engineer Associate	2024	• University of St. Gallen — Entrepreneurship	2024

TECHNICAL SKILLS

• Languages: Python, SQL, JavaScript/TypeScript, Java	• Data Engineering: ETL Pipelines, SQL Server, PostgreSQL, Python Automation
• ML & Deep Learning: Scikit-learn, TensorFlow, Keras, EfficientNet, Vision Transformers, LightGBM, XGBoost	• Databases & ORM: SQL Server, PostgreSQL, MySQL, Drizzle ORM
• Computer Vision: YOLOv8, OpenCV, SORT, CNNs, Transfer Learning	• BI & Analytics: Power BI, DAX, Power Query
• LLMs & Agents: MindStudio, Prompt Engineering, Agent Orchestration, RAG, LLM Evaluation	• Full-Stack: React 19, Express.js, Node.js, REST APIs
• MLOps & Deployment: Model/Agent Deployment, Evaluation & Monitoring, Git-based CI/CD, Streamlit	• Cloud & Tools: Microsoft Azure, AWS, GitHub, Git
	• ERP Systems: SAP Business One (SQL Server), Openbravo (PostgreSQL)